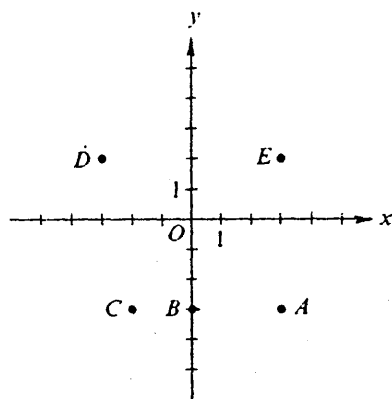


1. If $\frac{x}{3} + \frac{x}{3} = \frac{16}{3}$, then $x =$

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 16

2. If $x^3 = 21$, then, of the following, which is the closest approximation to x ?

- (A) 3
- (B) 7
- (C) 18
- (D) 63
- (E) 9,261



3. For which of the points in the figure above is the ratio of the x -coordinate to the y -coordinate equal to $\frac{3}{2}$?

- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

4. Marbles are drawn from a bag without replacement. First, 5 marbles are drawn, then 4, then 3, then 2, then 1 marble is drawn. From this point, the pattern of drawings is repeated 2 more times. After the drawings have been completed, 2 marbles remain in the bag. How many marbles were originally in the bag?

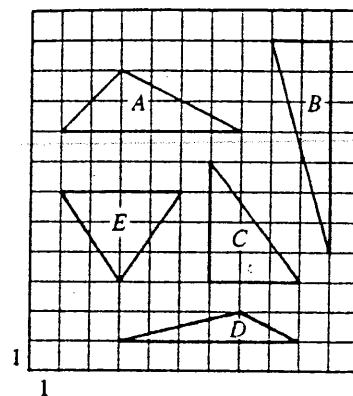
- (A) 43
- (B) 44
- (C) 47
- (D) 58
- (E) 62

5. $26^2 =$

- (A) 2×26
- (B) $2^2 \times 26$
- (C) $2^2 \times 6^2$
- (D) $2^2 \times 13^2$
- (E) 13×26

6. Jim charges \$3.50 an hour before midnight and \$5.00 an hour after midnight for baby-sitting. If he started baby-sitting at 6 p.m. and charged a total of \$28.50, at what time did he finish baby-sitting?

- (A) 12:30 a.m.
- (B) 1:00 a.m.
- (C) 1:30 a.m.
- (D) 2:00 a.m.
- (E) 2:30 a.m.



7. Which triangle on the grid above has the greatest area?

- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

8. All of the following are equal to 0.05 EXCEPT

- (A) $\frac{3}{40}$
- (B) $\frac{4}{80}$
- (C) $\frac{5}{100}$
- (D) $\frac{12}{240}$
- (E) $\frac{2\sqrt{2}}{40\sqrt{2}}$

9. What is the maximum number of square pieces, each measuring 2 inches by 2 inches, that can be cut from a rectangular piece of paper 16 inches by 14 inches?

- (A) 56
- (B) 64
- (C) 96
- (D) 112
- (E) 120

10. \$60 is 40 percent of

- (A) \$24
- (B) \$40
- (C) \$100
- (D) \$150
- (E) \$200

11. If $\frac{\frac{a}{3}}{\frac{b}{4}} = 1$, then $\frac{a}{b} =$

- (A) $\frac{3}{4}$
- (B) $\frac{4}{3}$
- (C) 3
- (D) 4
- (E) 12

12. If x is an even number, what is the next odd number greater than $3x + 1$?

- (A) $x + 1$
- (B) $x + 2$
- (C) $x + 3$
- (D) $3x + 2$
- (E) $3x + 3$

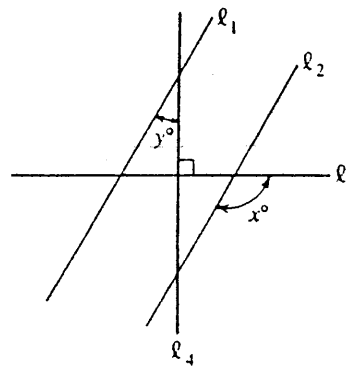
13. If x and y are integers, $x - y = 0$ and $y \neq 0$, which of the following must be true?

I. $x = 0$

II. $x = y$

III. $\frac{x}{y} = 1$

- (A) I only
- (B) II only
- (C) I and II
- (D) I and III
- (E) II and III



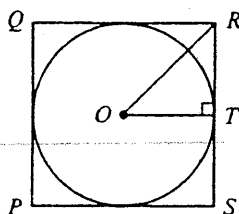
Note: Figure not drawn to scale.

14. In the figure above, if $l_1 \parallel l_2$, $l_3 \perp l_4$, and $x = 95$, what is the value of y ?

- (A) 2.5
- (B) 5
- (C) 10
- (D) 15
- (E) 20

15. At a certain high school, 45 percent of the first-year class take Spanish and 35 percent take history. How many first-year students are not enrolled in either class?

- (A) 10
- (B) 20
- (C) 25
- (D) 55
- (E) It cannot be determined from the information given.

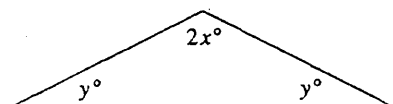


16. In the figure above, O is the center of the circle that is inscribed in square $PQRS$. If the area of $\triangle ORT$ is 8, what is the radius of the circle?

(A) 2
(B) 4
(C) 8
(D) $2\sqrt{2}$
(E) $4\sqrt{2}$

17. If w , x , y , and z are consecutive positive integers such that $w > x > y > z$, what is the product $(w - x)(x - y)(y - z)$?

(A) -1
(B) 0
(C) 1
(D) 3
(E) It cannot be determined from the information given.



18. In the triangle above, what is the value of y in terms of x ?

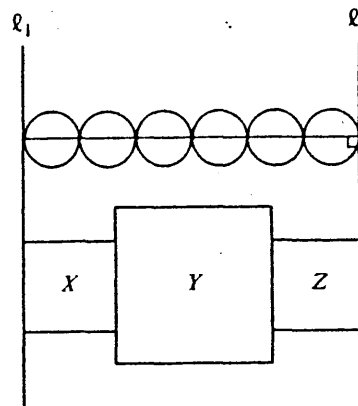
(A) $60 - x$
(B) $90 - x$
(C) $90 - 2x$
(D) $150 - 2x$
(E) $180 - x$

19. One side of a fair coin has the numeral 1 written on it and the other side has the numeral 2 written on it. If the coin is tossed 3 times and the numeral that lands up is recorded each time, what is the probability that the sum of the 3 tosses is at least 4?

(A) $\frac{1}{8}$
(B) $\frac{1}{4}$
(C) $\frac{1}{2}$
(D) $\frac{3}{4}$
(E) $\frac{7}{8}$

20. A student's average (arithmetic mean) on the first 4 tests in an algebra course was 75. After taking the fifth test, the student's average was 78. What was the student's grade on the fifth test?

(A) 90
(B) 88
(C) 86
(D) 81
(E) It cannot be determined from the information given.



21. In the figure above, $l_1 \parallel l_2$. The perimeters of squares X and Z are each 20. The sum of the circumferences of the 6 circles of equal size is 18π . What is the perimeter of square Y ?

(A) 28
(B) 30
(C) 32
(D) 36
(E) 40

22. If the difference between two numbers is greater than their sum, which of the following must be true about the numbers?

- (A) They are both positive.
 - (B) They are both negative.
 - (C) At least one of them is positive.
 - (D) At least one of them is negative.
 - (E) One of them is positive and the other is negative.
-

23. Let the relation $\lceil$ be defined for all pairs of numbers (a, b) and (s, t) as follows:

$(a, b) \lceil (s, t)$ is true if

(1) $a < s$, or if

(2) $a = s$ and $b < t$;

otherwise $(a, b) \lceil (s, t)$ is false.

For which of the following is $(a, b) \lceil (s, t)$ true?

I. $(3, 7) \lceil (3, 5)$

II. $(1, 9) \lceil (3, 5)$

III. $(8, 1) \lceil (3, 5)$

- (A) None
- (B) I only
- (C) II only
- (D) III only
- (E) II and III

24. If x and y are positive integers and $\frac{x}{3} < y < \frac{x}{2}$, what is the smallest possible value for x ?

- (A) 7
 - (B) 6
 - (C) 5
 - (D) 4
 - (E) 3
-

25. Suppose that t is equal to $p \times n$. If p is decreased by 25 percent while t is increased by 20 percent, by what percent must n be increased to maintain the original relationship between t , p , and n ?

- (A) $62\frac{1}{2}\%$
- (B) 60%
- (C) 45%
- (D) $37\frac{1}{2}\%$
- (E) 5%

1. An express train travels twice as fast as a local train, and the rate of the local train is r miles per hour. In terms of r , what is the rate of the express train in miles per hour?

(A) $r - 2$
 (B) $r + 2$
 (C) $2r$
 (D) r^2
 (E) $\frac{r}{2}$

2. If $x + y = 8$ and $3 + y = 4$, then $3 + x =$

(A) 10
 (B) 12
 (C) 15
 (D) 21
 (E) 27

3. When 20 is subtracted from 4 times a certain number, the result is 100. What is the number?

(A) 120
 (B) 80
 (C) 40
 (D) 30
 (E) 20

4. In a certain right triangle, the difference between the measures of the two smaller angles is 30 degrees. What is the degree measure of the smallest angle in this triangle?

(A) 15
 (B) 30
 (C) 45
 (D) 60
 (E) 75

5. When the sum of 1.343 and 2.834 is rounded to the nearest tenth, the result is

(A) 4.1
 (B) 4.17
 (C) 4.177
 (D) 4.18
 (E) 4.2

6. $\frac{5}{18}$ of what number is 10?

(A) 36
 (B) 9

(C) $\frac{9}{2}$
 (D) $\frac{25}{9}$
 (E) $\frac{1}{36}$

7. The number 777,777,777 is divisible by which of the following?

(A) 11
 (B) 77
 (C) 111
 (D) 1,111
 (E) 7,777

8. Points C and D lie on line segment AB . C is the midpoint of AB , and the length of CD is $\frac{1}{3}$ the length of CB . If the length of CD is 2, what is the length of AB ?

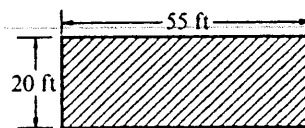
(A) 8
 (B) 10
 (C) 12
 (D) 16
 (E) 24

9. Arnie is now 8 years older than Bessie. In 5 years, Arnie will be twice as old as Bessie will be then. How many years old is Arnie now?

(A) 3
 (B) 5
 (C) 8
 (D) 11
 (E) 16

10. If $2x = 3y = 5z$ and x , y , and z are different positive integers, what is the least possible value of xyz ?

(A) 30
 (B) 60
 (C) 90
 (D) 150
 (E) 900



11. The shorter sides of the rectangular metal sheet above are to be joined without overlap to form a right circular cylinder with radius r feet. Of the following, which is the best approximation of r ?

(A) 3
 (B) 9
 (C) 14
 (D) 18
 (E) 27

12. If $n = 50 \times 40$, then $\frac{1}{n} =$

(A) 0.0002
 (B) 0.0005
 (C) 0.002
 (D) 0.005
 (E) 0.02

13. The average (arithmetic mean) of three numbers is 1 less than the average of the two numbers that remain after one number has been eliminated. If the eliminated number is 1, what is the average of the three numbers?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

14. If $7 + 3x = 7 + y$ and $2x = y$, which of the following must be true?

I. $7 + 3x > y$
II. $x = y$
III. $3x + y = 14$

(A) None
(B) I only
(C) III only
(D) I and II
(E) I and III

15. If $\frac{1}{2}$ of the length of the longer side of a rectangle equals $\frac{2}{3}$ of the length of the shorter side, what is the ratio of the length of the longer side to the length of the diagonal of the rectangle?

(A) $\frac{1}{3}$
(B) $\frac{2}{3}$
(C) $\frac{3}{4}$
(D) $\frac{3}{5}$
(E) $\frac{4}{5}$